

Isaac DeFrain

Senior Full-Stack Engineer - *blockchain & zero-knowledge*

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I have an insatiable passion for learning about and building open source, decentralized, and formally verified zero-knowledge blockchain protocols

University Education

Kent State University, Kent, Ohio

A.B.D. Pure Mathematics, 2018 - complex analysis & potential theory (published)
M.A. Pure Mathematics, 2012

Harriet L. Wilkes Honors College, Florida Atlantic University, Jupiter, FL

B.A. Mathematics and Physics, Minor: Chemistry, 2010
Thesis: The Alspach Norm in Classifying Complemented Subspaces of $L_p, p > 2$

Software Work Experience

LayerZero Labs (Ethereum, Starknet), Jun 2025 - Oct 2025

Senior Protocol Engineer

Implemented LayerZero omnichain messaging protocol in Cairo (Zellic/Ottersec audited)
Implemented Starknet wBTC and SolvBTC OFTs (Omnichain Fungible Tokens) in Cairo
Deployed and wired Solidity/Cairo messaging protocol endpoint contracts
End-to-end unit, integration, and fuzz tested Cairo endpoint contracts
Integrated Cursor into my workflow for increased productivity

Granola Systems (Mina Protocol), Dec 2022 - Jun 2025

Senior Software Developer/Security Engineer

Architected and implemented performant blockchain indexer for Mina Protocol (mina-indexer)
Created generic integration testing harness using docker compose (abstract-engine/minimina)
Fixed critical go-libp2p go-mplex DDoS vulnerability without requiring hard fork (go-mplex)
Led security/QA investigations targeting release of Mina zkApps update (zkapps-itn)
Added local precomputed block logging to Mina daemon supporting Mina Indexer (mina)

Viable Systems (Tezos), Dec 2020 - Mar 2022

Formal Verification Engineer

TLA+ spec'd Emmy/Tenderbake consensus, P2P protocols, task schedulers (specs)
Translated TLA+ specs into rust implementations for TezEdge (Tezos rust node)
Built liveness-checking and node-debugger proofs of concept

RChain Cooperative (RChain), Dec 2018 - Mar 2020, Jul 2022 - Sep 2022

Smart Contract/Formal Verification Engineer

Implemented PoS consensus and dynamically dispatchable smart contracts (rchain)
K Framework spec'd and verified Rholang smart contract language (rho)
TLA+ spec'd CBC Casper consensus (casper)

	Wyzant, Mar 2020 - Dec 2020 & Sep 2022 - Dec 2022 <i>Computer Science, Programming, Cryptography, Solidity Tutor</i> Provided excellent one-on-one online learning sessions in many areas including: programming, smart contracts, applied cryptography (graduate level), theory of computation, dependent types
Personal Projects	<i>scotus-helper</i>: RAG infra/chatbot for SCOTUS opinions https://scotus-helper.com (2026) <i>blog</i>: Blog reader web app and personal blog https://blog.isaacdefrain.com (2025) <i>github-reporting-py</i>: GitHub commit CSV reporting tool for invoicing (2023) <i>mina-indexer-block-util</i>: <i>gsutil</i> wrapper for efficiently handling Mina blocks for indexing (2023) <i>model-based trace-checking</i>: check conformance of implementations to specifications (2022) <i>bft-specs</i>: Byzantine fault tolerant TLA+ specifications (2022) <i>TLAplusFun</i>: explorations with TLA+ specifications and Apalache model checker (2020 - 2022)
Skills	Rust, OCaml, Solidity, Cairo, TLA+, TypeScript, Python, Bash, Ruby, Linux (debian), nixOS, Docker, CI/CD, AI, EVM/non-EVMs, testing, debugging, clear communication, collaboration
Independent Study	ByteByteAI: Building with Claude Code - cohort 1, May 2026 Parsity: Applied AI Accelerator - cohort 2, Apr - May 2026 Scalable, transparent, and post-quantum secure computational integrity, Feb 2026 RISC Zero zkVM: Scalable, Transparent Arguments of RISC-V Integrity, Jan 2026 Scalable and Probabilistic Leaderless BFT Consensus through Metastability, Jan 2026 PlonK, Jan 2026 Proofs, Arguments, and Zero-Knowledge, Dec 2025 - present Jolt Book, Dec 2025 - present ZK Hack V, Nov - Dec 2024
Developer Education	TLA+ Workshops <i>Viable Systems, Jan 2021 - Mar 2022</i> Organized and led weekly TLA+ workshops to improve our developers' abilities to understand and write TLA+ specifications Blockchain and Rholang Workshop <i>Blockchain AI Center, Gimcheon, South Korea, Dec 10-14, 2018</i> Prepared and taught 5-day workshop covering everything from PoW/PoS and computational calculi basics to writing and depolying Rholang smart contracts using Cryptofex and RNode RChain Cooperative <i>Technical Consultant, August 2018 - November 2018</i> Technical blog writing explaining mathematical foundations of RChain's blockchain protocol
Publications	Chebyshev polynomials associated with a system of continua <i>New Trends in Approximation Theory, Fields Institute Communications 81, Springer-Verlag 2018</i>
Conference Talks	TLA+ Formal Specification of Bootstrapping <i>Nomadic Labs Research Seminar #15 (Tezos), May 25, 2021</i> Chebyshev polynomials on a continuum in the complex plane <i>Northeastern Analysis Meeting, SUNY Brockport, October 2016</i> Chebyshev polynomials on a system of continua <i>New Trends in Approximation Theory, The Fields Institute, Toronto, ON, July 2016</i> Chebyshev polynomials on a system of continua <i>15th International Conference on Approximation Theory</i> <i>Menger Hotel, San Antonio, TX, May 2016</i>

Mathematics Research	Mathematical Sciences, Kent State University , Fall 2013 - Spring 2018 Advisor: Dr. Artem Zvavitch, 2017-2018 Project: Almost Isometric Embeddings of Finite Metric Spaces
	Advisor: Dr. Volodymyr Andreievskii, 2015 - 2017 Project: Potential Theory and Chebyshev Polynomials on Quasiconformal Domains
	Advisor: Dr. Fedor Nazarov, 2013 - 2015 Project: Conformal Invariants and Geometric Function Theory, 2014-2015 Project: Hörmander Existence Theorem and Plurisubharmonic Functions, 2013-2014
Teaching Experience	Mathematical Sciences, Kent State University Instructor, Analytic Geometry and Calculus II, Fall 2017 Instructor, Analytic Geometry and Calculus II, Spring 2017 Instructor, Modeling Algebra, Fall 2016 Instructor, Analytic Geometry and Calculus I, Spring 2016 Instructor, Modeling Algebra, Fall 2015 Instructor, Analytic Geometry and Calculus II, Spring 2014 Instructor, Analytic Geometry and Calculus I, Fall 2013 Instructor, Analytic Geometry and Calculus I, Spring 2013 Instructor, Calculus for Life Sciences, Fall 2012
Mentoring	Undergraduate Research Advisor, Choose Ohio First Scholarship Spring 2018: The Geometry of the Power Rule (<i>Award of Excellence</i>) Spring 2012: Applications of the Golden Ratio Spring 2011: X-ray Crystallography & the Fourier Transform
	Undergraduate Research Experience, Kent State University Graduate student mentor, Summer 2015
Workshops	
	Mathematical Sciences Research Institute, Berkley, CA <i>Summer Graduate School - Mathematical General Relativity</i> , July 2012
Volunteer Work	
	Category Theory & Functional Programming Collaborative Learning Coordinator <i>Group to discuss category theory and functional programming</i> Check out our GitHub repo
References	
	Upon request